February 16, 2026 5:16 PM

Carboxylic Acid:

↳ Weak acids and are found in citrus fruits, crabapple, rhuburb, and other foods

↳ lactic acid is found in sour milk (ew) and yogurt. When your muscles hurt after working out, it's because there is lactic acid in them.

↳ Contain a carboxyl group ($-\overset{O}{\overset{\|}{C}}-OH$)

↳ combines a hydroxyl ($-OH$) with a carbonyl ($-\underset{O}{\underset{\|}{C}}-$)

Properties

↳ Properties of Carboxylic acids

↳ The carboxl bond ($-\overset{O}{\overset{\|}{C}}-OH$) is polar and can form hydrogen bonds.

↳ 1-4 carbon atoms = soluble as alcohols
Larger ones = insoluble

↳ Have the same properties as acids and can be distinguished using a litmus paper test. can also be neutralized with organic bonds to form salts.

↳ Higher melting points than other hydrocarbons

Naming

↳ ① Find the longest chain containing the carboxyl group
remove the -e, and add the -oic acid.

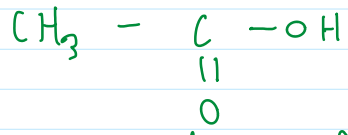
② give the lowest number priority to the carboxylic acids.

Common carboxylic acids:

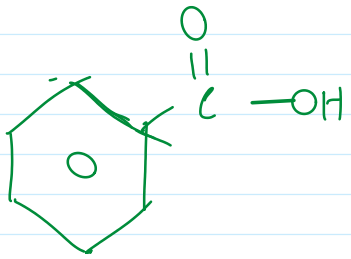
① Methanoic acid ($HCOOH$) → formic acid

$$H-\overset{O}{\overset{\|}{C}}-OH$$

2) Ethanoic acid → acetic acid.



3) Phenylmethanoic acid → Benzoic acid.



Reactions

Alcohol + (O) → aldehyde + water (H_2O)
oxidation

Aldehyde + (O) → Carboxylic acid
oxidation

★ Ketones can't be used, since there are no H atoms for the oxygen to get with.

The (O) is provided by chromate ion in it's Cr^{+6} oxidation phase. Orange color in aq. Used in breathalyzer tests.

↳ A set amount of air containing ethanol is changed from an alcohol, to a aldehyde (Ethanal) to a carboxylic acid (Ethanoic acid). This turns Cr^{6+} (orange in aq) to Cr^{3+} (green in aq). How green it is tells the alcohol concentration.